Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/29/2025
PI Name:	Zea, Luis Ph.D.		
Project Title:	Multi-Generational Genome-Wide Yeast Fitness Profiling Beyond and Below Earth's van Allen Belts		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Luis.Zea@Colorado.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	407-242-2885
Organization Name:	University of Colorado, Boulder		
PI Address 1:	Aerospace Engineering Sciences		
PI Address 2:	429 Ucb, ECAE 1B02		
PI Web Page:			
City:	Boulder	**State:**	CO
Zip Code:	80309-0429	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-Artemis1 (EM1). App A: Orion (Artemis-1) (formerly Exploration Mission-1)
Start Date:	05/01/2019	**End Date:**	12/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	9	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
Contact Email:	Denise.E.Freeland@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 5/23/24) NOTE: End date changed to 04/29/2024 per NSSC information (Ed., 5/31/23) NOTE: End date changed to 04/30/2023 per NSSC information (Ed., 4/12/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Stodieck, Louis Ph.D. (University of Colorado, Boulder) Nislow, Corey Ph.D. (University of British Columbia, Canada)		
Grant/Contract No.:	80NSSC19K0708		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	As human space exploration expands beyond lower Earth orbit, it is necessary to characterize the effects of space radiation, microgravity, and the combination thereof on cells. Because it is complicated to have large sample numbers when studying the effects of different factors on humans, scientists commonly use model organisms that share some of the key aspects being studied. In this case, we will use yeast, as around 70% of its essential genes have a significant human homolog. More specifically, this project will use a molecularly barcoded yeast genome-wide knockdown collection that will enable the systematic interrogation of the effect of microgravity, space radiation, and a combination thereof in each gene. Each strain in the collection has a single gene deleted and a representative molecular barcode that enables quantifying the fitness of each mutant under the test conditions, by measuring the relative abundance at different points in time. To differentiate the effects of microgravity and space radiation on each strain, an experimental set will be flown beyond the van Allen belts on Orion's Exploration Mission 1 (EM-1) (considered in microgravity and irradiated by space radiation) and equivalent sets will be cultured asynchronously on board the International Space Station (ISS) (considered in microgravity but mostly – although not completely – protected of space radiation by the van Allen belts) in our smart incubator (Space Automated Bioproduct Lab (SABL)) and on Earth (also in a ground SABL). Each of the ISS and Earth experiments will include two sets: one where the temperature profile experienced during the EM-1 flight is replicated, and a second cultured at a constant temperature to determine the potential role of temperature variation on the results from EM-1. The first aim of this project is to identify the metabolic and genomic pathways in yeast affected by microgravity, space radiation, and a combination of both. The second one is to differentiate between gravity and radiation exposure on single-gene deletion mutants' ability to thrive in the spaceflight environment. We hypothesize that mutants lacking genes associated with DNA repair, recombination, and replication will have lower survivability rates beyond the van Allen belts than their below van Allen belts- or Earth-controls The experiment is designed to have a controlled start after Orion is past the van Allen belts, grow ~21 generations of the deletion series, and fix or preserve samples for post-flight analyses. Should the automated controlled approach be considered inappropriate for implementation on EM-1, we have a passive approach that is based on dotting each mutant individually on agar. We have performed both approaches in space in the past. This project will address three Space Biology Program Science Elements, three Objectives, three Guiding Questions, and four Decadal Survey's highest priority Recommendations by preserving nucleic acids of different generations of the yeast deletion series cultures grown in space, beyond as well as below the van Allen belts (and uploading the genomic and transcriptomic data to GeneLab).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project integrates data on the molecular and cellular mechanism of radiation damage, which can serve to improve prediction of risk of cancer as a function of radiation dosage and to evaluate the effectiveness of potential countermeasures.
Task Progress:	The main results, conclusions, methodology, and potential next steps are presented in our manuscript "One Small Step for Yeast — NASA's Artemis I mission reveals Lunar spaceflight induces DNA methylation regulating transcription in yeast", under review. Additionally, the product and value generated from this project is documented as follows: Books 1. Pathak, Y., Araújo dos Santos, M., Zea, L. Handbook of Space Pharmaceuticals. Springer International Publishing Peer-Reviewed Journal Papers 2. Gaikani, et al., One Small Step for Yeast — NASA's Artemis I mission reveals Lunar Spaceflight induces DNA methylation regulating transcription in yeast, under review, Cell 3. Zea, L., Piper, S. S., Gaikani, H., Khoshnoodi, M., Niederwieser, T., Hoehn, A., ... & Nislow, C. (2022). Experiment verification test of the Artemis I 'Deep Space Radiation Genomics' experiment. Acta Astronautica, 198, 702-706. https:// Conference Papers and Posters, Open-Access Code, Podcasts, and STEM outreach 4. Zea, L., Levine, H., Santamaria, S., Hammond, T., Brandizzi, F., Henessa, T., Artemis I Biological Payloads Panel, ASGSR 2023 5. Zea, L., Nislow, C., Artemis I's Deep Space Radiation Genomics Opportunities, Constraints, & Challenge Solving, NASA Artificial Intelligence / Machine Learning (AI/ML) Working Group, April 10, 2023 6. Zea, L., Niederwieser, T., Hoehn, A., Grusin, M., Wright, J., Stanish, G., Piper, S., Flores, P., Griffith, R., Gaikani, H., Khoshnoodi, M., Siems, K., Cortesao, M., Wahl, K,. Bernstein, V., Knipp, D., Wilson, K., Scott, K., Lutsic, A., Soler, M., Sato, K., Viau, A., Aragon, R., Stodieck. L., Countryman, S., Carr, C.E., Moeller, R., Nislow, C., From Idea to Lunar Orbit Flight Readiness - Implementation of the Artemis I 'Deep Space Radiation Genomics' (DSRG) Yeast Experiment, American Society for Gravitational and Space Research (ASGSR) Conference, November 3-7, 2021 7. Zea, L., Piper, S., Gaikani, H., Khoshnoodi, M., Niederwieser, T., Hoehn, A,. Grusin, M., Wright, J., Flores, P., Wilson, K., Lutsic, A., Stodieck, L., Carr, C.E., Moeller, R., Nislow, C., Experiment Verification Test of the Artemis I 'Deep Space Radiation Genomics' Experiment, IAC-21-A.2.7x65012, 72th International Astronautical Congress, Dubai, United Arab Emirates 8. Zea, L., Nislow, C., First Controlled Biological Experiments beyond the Van Allen Belts, American Institute of Aeronautics and Astronautics (AIAA) Rocky Mountain Chapter Annual Technical Symposium, University of Colorado Boulder, September 29, 2021 9. Bernstein, V. and Wahl, K. (2021). Identifying and Visualizing the Moon's Position with respect to the Earth's Magnetotail (v1.0.0). Zenodo. [Github] 10. Wahl, K.R., Bernstein, V., Knipp, D., and Zea, L., Maximizing Space Radiation Exposure for the Deep Space Radiation Genomics (DSRG) Experiment on Artemis 1, 18th Space Weather Conference in the American Meteorological Society 101st Meeting 10-14 January 2021 (Virtual) 11. Fiske Planetarium Podcast "A View From Earth", Episode 8 - Life finds a Way - Or does it? Guests: Luis Zea & Bruce Jakosky, July 2020 [Podcast] [Video] 12. Wahl, K., Bernstein, V., Knipp, D. Deep Space Radiation Genomics (DSRG) Experiment on Artemis 1: Magnetotail and Bow shock Affect, Research Experience for Undergraduates (REU) Final Presentations, University of Colorado, Boulder, July 29, 2020 13. Meda, E., Herbruger, E., Hernandez, D., Aragón, R., Aguirre, D., Alvarado, I., Zea, L., Viau, A., Piezas para proyecto DSRG - Artemis 1, UVG 2020 Science Fair 14. Knipp, D., Wahl, K., Bernstein, V., Zea, L. Sending Tiny Life Forms to Space. ED032 - The Up-Goer Five Challenge: Explaining big important things in simple words. AGU 2020 Fall Meeting, December 1-7, 2020 15. Breaux S, Fuchs F, Cortesao M, Siems K, Stodieck L, Niederwieser T, Zea L, Carr CE, Moeller R, Nislow C. "Irradiation Ground Control for a Genome-Wide Yeast Fitness Profiling Experiment On Board Orion's Artemis 1 Mission." 35th Annual Meeting of the American Society for

	Gravitational and Space Research, Denver, CO, November 20-23, 2019. 16. Zea L, Niederwieser T, Stodieck L, Carr C, Moeller R, Nislow C. "Experiment Design for a Genome-Wide Yeast Fitness Profiling Experiment On Board Orion's Artemis 1 Mission." 70th International Astronautical Congress (IAC), Washington, DC, October 21-25, 2019. 17. Zea, L., Niederwieser, T., Stodieck, L,. Carr, C., Moeller, R., Nislow, C., Experiment Design for a Genome-Wide Yeast Fitness Profiling Experiment On Board Orion's Artemis 1 Mission, IAC-19-A2.7.9x51501, 70thInternational Astronautical Congress (IAC) 2019, Washington, D.C.
Bibliography Type:	Description: (Last Updated: 11/13/2025)